Notes: Bennett, Bettis, Gopalan, and Milbourn (JFE, 2017)

Compensation goals and firm performance

Contents

1 What this paper is doing

- The paper studies performance goals embedded in executive incentive contracts (threshold/target/maximum) and asks whether firms' *reported* performance clusters just above those goals.
- Empirical signature: a discontinuity in the density of Actual−Goal at zero (too many observations just above 0 relative to just below 0).
- Main finding: there is a strong bunching just above compensation goals, especially for EPS goals, for single-metric plans, for concave pay-performance relationships around targets, and for non-equity (cash) payouts.
- Mechanisms explored: (i) target ratcheting (better performance today raises future targets), (ii) forced turnover risk if targets are missed, (iii) (hard-to-test) reference-dependent preferences.
- How firms beat goals: evidence consistent with accrual manipulation and cuts to discretionary spending (R&D, SG&A) to just exceed targets.

2 Data and sample construction

- Performance-goal / contract data: Incentive Lab (IL), sourced from proxies.
- Universe: grants to named executives among the 750 largest firms by market cap, 1998–2012; much analysis focuses on 2006–2012 (post SEC standardized disclosure).
- Main analysis sample: CEO grants tied to *absolute accounting* metrics matchable to Compustat:

  EPS, Earnings, Sales, EBIT, EBITDA, EBT, Operating income, FFO.

- Final main sample: 974 firms, 5,810 CEO grants (2006–2012).

3 Key variables and notation

- Let $G_{i,t}$ be the stated goal in the contract (threshold or target), and $A_{i,t}$ be Compustat actual performance.
- For a metric m (EPS / Sales / Profit), the core running variable is:

$$x_{i,t}^{(m)} = A_{i,t}^{(m)} - G_{i,t}^{(m)}.$$

- They work with:
  - Actual less target EPS $= A^{EPS} - G^{EPS}_{target}$ (with Compustat EPS definition chosen as the one closest to the disclosed target).
  - Actual less target sales $= \dfrac{A^{Sales} - G^{Sales}_{target}}{\text{Total Assets}}$.
  - Actual less target profit $= \dfrac{A^{Profit} - G^{Profit}_{target}}{\text{Total Assets}}$.

- Combined measure used for power:

$$\text{Actual less target} = \frac{x^{EPS}}{\sigma(x^{EPS})} \cup \frac{x^{Sales}}{\sigma(x^{Sales})} \cup \frac{x^{Profit}}{\sigma(x^{Profit})}$$

  i.e., they standardize each metric-specific difference by its SD and pool them.

- Concave vs. convex at target (classification rule):

$$\text{Concave if } \frac{\text{slope left of target}}{\text{slope right of target}} > 1.01, \qquad \text{Convex if } \frac{\text{slope left}}{\text{slope right}} < 0.99,$$

  else linear.

- Single- vs multiple-metric: "Multiple" if more than 50% of grant payout depends on more than one metric.

4 All models / empirical tests used in the paper (as equations/algorithms)

4.1 Density discontinuity test: McCrary (2008) / `DCdensity`

Goal: test whether the density of x jumps at $x = 0$ (the goal).

- Step 1: build a finely binned histogram of x, with bins chosen so no bin straddles 0.
- Default bin width used by `DCdensity`:

$$b = 2\sigma n^{-1/2},$$

  where σ is sample SD and n is sample size.

- Step 2: smooth the histogram by local regressions on each side of 0 using a triangular kernel.
- Default bandwidth (as described in the paper) is computed on each side and then averaged. For a side-specific regression:

$$h = 3.348 \left[\frac{\tilde{\sigma}^2 (b-a)}{\tilde{f}(X_j)^2} \right]^{1/5},$$

  where $\tilde{\sigma}^2$ is the regression MSE, $b - a$ is a side-specific range term (right side uses X_j, left uses $-X_j$), and $\tilde{f}(X_j)^2$ is the estimated second derivative implied by a global polynomial model.

- Hypothesis test: Wald test of $\Delta \hat{f}(0) = \hat{f}_+(0) - \hat{f}_-(0) = 0$.

4.2 Discontinuity test across the whole density: Bollen & Pool (2009) style

- Histogram bin width chosen to minimize MSE:

$$b = 0.7764 \times 1.364 \times \min\left[\sigma, \frac{Q}{1.34}\right] \times n^{-1/5},$$

  where Q is the IQR.

- Smooth density via Gaussian kernel with bandwidth set equal to the bin width.
- Compare actual bin counts to expected bin counts implied by the smooth density; compute a t-statistic for each bin based on sampling variation (Demoivre–Laplace / binomial approximation).

4.3 Bootstrap "just above vs just below" test

- For a given bin width b, repeatedly draw a random sample (e.g., 50 or 100 obs) and compute:

$$D = \#\{x \in (0, b]\} - \#\{x \in [-b, 0)\}.$$

- Repeat 1,000 times; test whether $E[D] > 0$, and in cross-sections compare $E[D]$ across subsamples.

4.4 Regression-based density/bunching model (Equation (1) in the paper)

They create bins in *performance space* for EPS, Sales, and Profit separately (bin size per McCrary) and then pool bins (with metric interactions) to estimate whether goals predict bunching of realized performance in the same bin.

$$\begin{aligned}\text{Number of firms}_{b,m,t} = \alpha &+ \beta_0\,(\text{Metric}_m \times \text{Midpoint}_b) + \beta_1\,(\text{Metric}_m \times \text{Midpoint}_b^2) \\ &+ \beta_2\,(\text{Metric}_m \times \text{Midpoint}_b^3) + \beta_3\,(\text{Metric}_m \times \text{Midpoint}_b^4) + \beta_4\,\text{Number of goals}_{b,m,t} + \Upsilon.\end{aligned} \tag{1}$$

where:

- Number of firms = log(1 + #firms with realized performance in bin b).
- Number of goals = log(1 + #firms with threshold/target goals in bin b).
- Υ includes year fixed effects; errors clustered at the bin level (as described).

4.5 Target ratcheting tests (Table 3, Panel A and Panel B)

Panel A (goal-setting persistence):

$$G_{i,t}^{(m)} = \alpha + \beta\,A_{i,t-1}^{(m)} + \text{controls} + \delta_{\text{industry}\times t} + \varepsilon_{i,t}.$$

Panel B (meeting next target):

$$\text{MeetTarget}_{i,t+1} = \alpha + \beta\,\text{ExceedTarget}_{i,t} + \text{controls} + \delta_{\text{industry}\times t} + \varepsilon_{i,t}.$$

(ExceedTarget is 1 if performance is in the two bins just above the target in t.)

4.6 Forced CEO turnover test (Table 4)

Linear probability style regression (as presented in table form):

$$\text{Forced}_{i,t+1} = \alpha + \beta\,\text{MissTarget}_{i,t} + \theta_1\text{Target}_{i,t} + \theta_2\text{Target}_{i,t}^2 + \text{controls} + \delta_{\text{industry}\times t} + \varepsilon_{i,t}.$$

4.7 How firms beat goals (Equation (2) in the paper)

$$y_i = \alpha + \beta_0\,\text{Exceed EPS/Sales/Profit}_i + \beta_1\,\text{Size}_i + \beta_2\,\text{Market-to-book}_i + \Upsilon + \gamma_j + \varepsilon_i, \tag{2}$$

where $y_i \in \{\text{Accruals}, \Delta R\&D, \Delta SG\&A, \text{Repurchase}\}$, and γ_j are industry fixed effects (2-digit SIC) with year fixed effects included; SE clustered at firm level (as described).

5 Interpretation of all figures

Figure 1: Actual less target (combined EPS/Sales/Profit, standardized)

- Panel A: histogram of pooled Actual − Target is centered near 0 but visibly asymmetric: more mass just above 0 than just below.
- Panel B (McCrary): clear discontinuity at 0 with $p = 0.01$. Interpretation: too many firm-years barely exceed the target relative to barely missing it.
- Panel C (Bollen & Pool style): t-statistics across the distribution show the strongest spike around 0 (above the 99% threshold) and particularly on the right side of 0. Interpretation: the density is not smooth at the target; the "excess mass" is concentrated just above the goal.

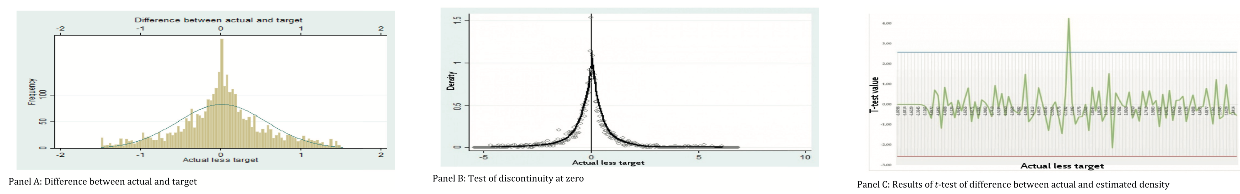


Figure 1: Difference between actual and target: EPS/Sales/Profit.

Figure 2: Metric-by-metric (EPS vs Sales vs Profit) at target

- Panel A (EPS): McCrary discontinuity is statistically significant at 0 (paper reports $p \approx 0.01$). The bunching is strongest here.
- Panel B (Sales): discontinuity not significant in McCrary test (paper reports $p \approx 0.41$), though there is still a visible spike.
- Panel C (Profit): discontinuity not significant in McCrary test (paper reports $p \approx 0.46$).
- Bootstrap evidence (reported in text): excess right-bin vs left-bin is much larger for EPS than for sales/profit (economic magnitude ordering: EPS $\gg$ Sales $\gtrsim$ Profit).

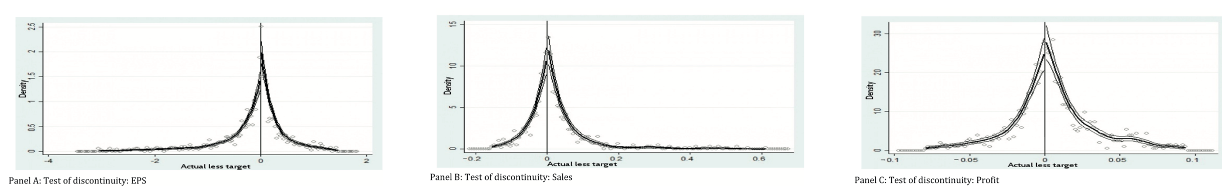
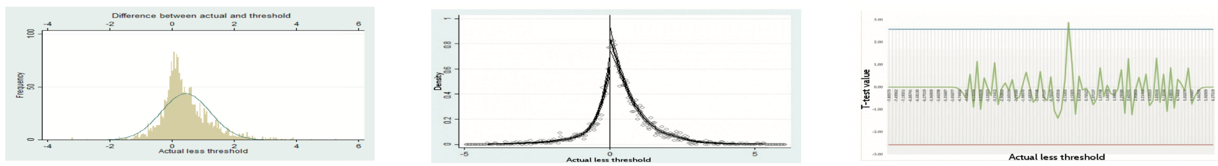


Figure 2: Difference between actual and target: EPS/Sales/Profit.

Figure 3: Actual less threshold (combined EPS/Sales/Profit, standardized)

- Panel A: histogram around threshold shows bunching around 0 and more mass right of 0.
- Panel B: McCrary indicates a significant discontinuity at 0 (paper reports $p \approx 0.018$).

- Panel C: the across-density t-test again shows a spike around 0.
- Interpretation: firms are also disproportionately likely to *just clear thresholds*, consistent with strong pay discontinuities at threshold.



Panel A: Difference between actual and threshold

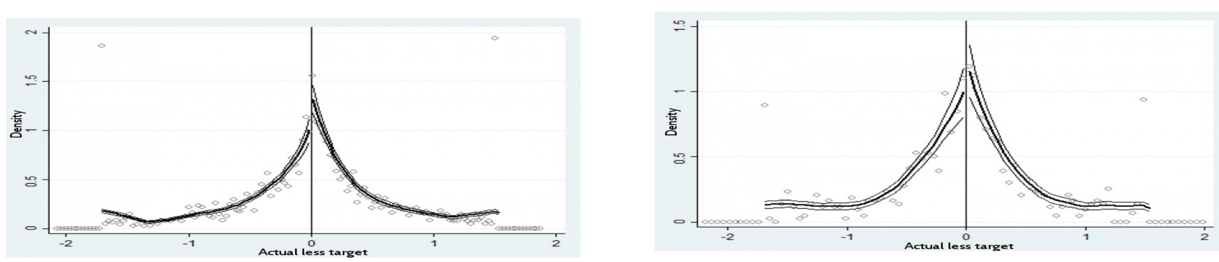
Panel B: Test of discontinuity at zero

Panel C: Results of *t*-test of difference between actual and estimated density

Figure 3: Difference between actual and threshold: EPS/Sales/Profit.

Figure 4: Single metric vs multiple metrics (at target, combined)

- Panel A (single metric): strong discontinuity at 0 (paper reports $p \approx 0.01$).
- Panel B (multiple metrics): essentially no discontinuity (paper reports $p \approx 0.90$).
- Interpretation: it is hard to "barely beat" multiple (positively correlated) goals simultaneously, so bunching is mainly in single-metric plans.



Panel A: *Actual less target* – Single goal

Panel B: *Actual less target* – Multiple goals

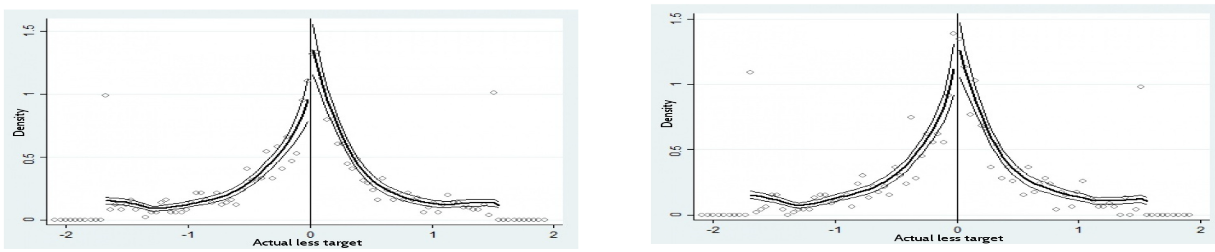
Figure 4: Difference between actual and target performance: single vs. multiple goals.

Figure 5: Concave vs convex pay-performance around the target

- Panel A (concave): discontinuity present (paper reports $p \approx 0.06$).
- Panel B (convex): weaker / not significant (paper reports $p \approx 0.11$).
- Interpretation: concavity makes the marginal benefit of outperforming beyond the target smaller, so managers have incentives to aim for "just above target" rather than "far above target."

Figure 6: Non-equity vs equity payout

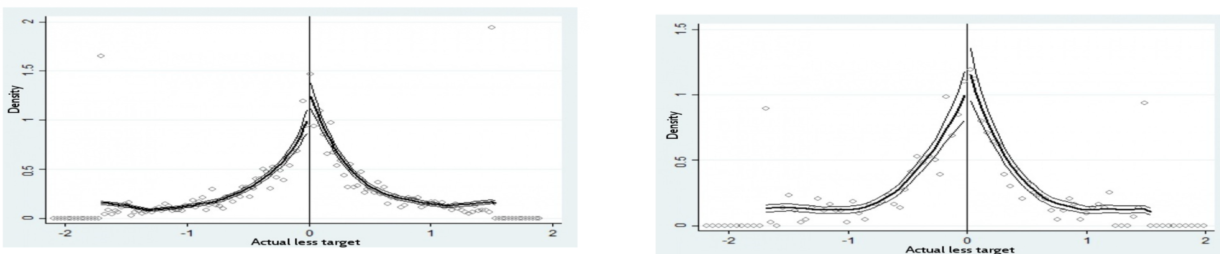
- Panel A (non-equity/cash): discontinuity at 0 (paper reports $p \approx 0.05$).
- Panel B (equity): no significant discontinuity (paper reports $p \approx 0.46$).
- Interpretation: bunching is strongest when payouts are non-equity (cash-like), consistent with (i) less convexity and (ii) more explicit "goal-based" payout incentives.



Panel A: *Actual less target* – Concave

Actual less target – Convex

Figure 5: Difference between actual and target performance: Concave vs. convex goals.



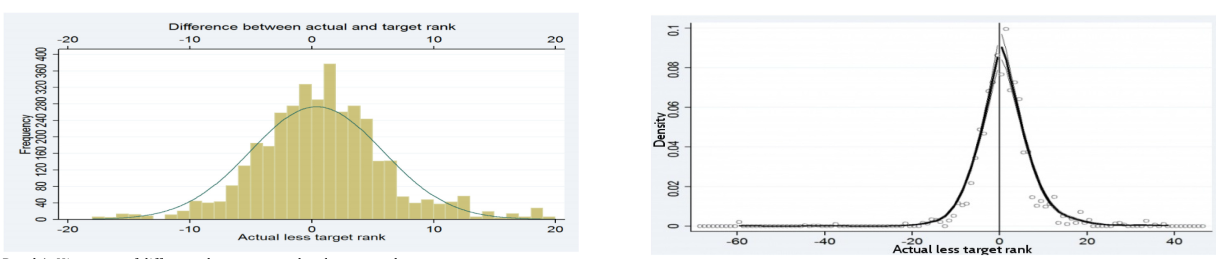
Panel A: *Actual less target* – Non-equity payouts

Panel B: *Actual less target* – Equity payouts

Figure 6: Difference between actual and target performance: non-equity vs. equity payouts.

Figure 7: Relative performance targets (rank/percentile) placebo

- Panel A: histogram of Actual rank − Target rank shows no excess mass just above 0; if anything, more firms just miss than just beat.
- Panel B: McCrary test finds no discontinuity at 0.
- Interpretation: relative-performance goals are harder to game via accounting manipulation (peer outcomes unknown until later), so the "just beat the target" pattern disappears. This strengthens the manipulation/goal-gaming interpretation for absolute targets.



Panel A: Histogram of difference between actual and target rank

Panel B: Test of discontinuity at zero of the difference between actual and target rank

Figure 7: Difference between actual and target ranks for relative performance grants.

6 All tables + interpretation of each

Interpretation of Table 1.

- EPS goals are the dominant accounting goal: 2,650 of 5,810 grants (about 46%).

Table 1. Summary characteristics

Panel A: Summary grant characteristics

	EPS	Earnings	Sales	EBIT	EBITDA	EBT	FFO	Operating income	Total
Number of CEOs/firms	492	291	456	106	224	135	35	399	974
Number of grants	2,650	882	1,910	294	751	347	147	1,466	5,810
Percent of grants that involve:									
Non-equity-based payout	0.295	0.129	0.253	0.045	0.114	0.054	0.020	0.228	0.723
Equity-based payout	0.120	0.031	0.069	0.007	0.023	0.011	0.005	0.048	0.285
Option payout	0.007	0.001	0.002	0.001	0.002	0.000	0.000	0.000	0.012
Long-term vesting	0.236	0.111	0.136	0.106	0.106	0.100	0.092	0.119	0.159
Convex	0.510	0.509	0.504	0.504	0.423	0.560	0.357	0.468	0.492
Concave	0.310	0.329	0.344	0.346	0.451	0.321	0.262	0.384	0.346
Linear	0.180	0.161	0.152	0.150	0.125	0.119	0.381	0.148	0.162
Multiple	0.199	0.254	0.345	0.178	0.201	0.257	0.125	0.256	0.223

Panel B: Performance goals and actual performance

Variable	N	Mean	SD	P25	Median	P75
Actual less target	3,805	−0.015	1.030	−0.357	0.011	0.346
Actual less threshold	2,357	0.389	1.029	−0.054	0.281	0.861
Actual less target EPS	1,606	−0.133	0.656	−0.300	0.010	0.150
Actual less threshold EPS	1,053	0.104	0.743	−0.127	0.101	0.450
Actual less target sales	948	0.025	0.097	−0.018	0.006	0.043
Actual less threshold sales	533	0.081	0.120	0.009	0.047	0.109
Actual less target profit	1,251	0.005	0.029	−0.009	0.003	0.017
Actual less threshold profit	771	0.015	0.029	−0.002	0.012	0.032

- Cash/non-equity payouts are most common (0.723 of grants involve some cash payout), which matters because the bunching results are strongest for non-equity.
- Only 15.9% are long-term vesting by their definition (final vesting $>$ 11 months after grant date).
- A large share of plans are convex (0.492) and a sizeable share are concave (0.346); the paper's discontinuity/bunching is more associated with concave structures.
- Actual performance is, in the median, very close to target (e.g., median Actual less target EPS = 0.01). This is exactly the setting where "just-barely beat" behavior could show up in density tests.

Interpretation of Table 2.

- Column (1): $\beta = 0.203$ on $\log(1+\#$ goals in bin$)$ means bins containing disclosed goals have systematically higher realized-performance mass. Roughly, $\exp(0.203) - 1 \approx 22.5\%$ higher expected count (paper summarizes as $\sim 20\%$).
- Column (2): both "single metric" and "multiple metrics" goal counts predict bunching in realized performance. Interestingly, the multiple-metric coefficient (0.212) exceeds the single-metric coefficient (0.142), which differs from the discontinuity-at-zero logic. I read this as: this regression detects *closeness* of realized performance to goals in a bin (not necessarily "just above"), and it also pools threshold/target goals.
- Column (3): only concave goals significantly predict bunching (0.275** vs 0.114 not significant). This matches the paper's narrative that concavity reduces incentives to outperform far beyond target and may induce "aim just above target" behavior.

Table 2. Reported performance and earnings-based pay targets

	Dependent variable: log(1 + # firms in bin)			
	(1)	(2)	(3)	(4)
Number of goals	0.203**			
	(0.082)			
Single metric		0.142**		
		(0.061)		
Multiple metrics		0.212**		
		(0.108)		
Concave			0.275**	
			(0.122)	
Convex			0.114	
			(0.092)	
Non-equity				0.146*
				(0.078)
Equity				0.478***
				(0.102)
Const.	2.829***	2.917***	2.875***	2.862***
	(0.104)	(0.104)	(0.102)	(0.102)
Obs.	11,576	11,576	11,576	11,576
R^2	0.668	0.676	0.637	0.674

- Column (4): both payout types show clustering in this regression, but equity is much larger (0.478***). Again, this regression is about being *near* any goal in a bin, not strictly a discontinuity just above the goal. So it can be consistent with Fig. 6 still showing discontinuity only for non-equity payouts.

Interpretation of Table 3.

- Panel A: targets are strongly positively related to lag performance for EPS, sales, and profit. This supports a "ratcheting" environment: good performance today leads to higher future goals.
- Panel B: firms that *just* exceed targets are about 9–11 percentage points more likely to meet next period targets. I read this as consistent with target ratcheting/goal dynamics: being near the target is predictive of being near (and meeting) next targets, potentially because boards set next targets based on past performance and/or because firms keep managing to land just above.

Interpretation of Table 4.

- Missing the target increases the probability of a forced CEO turnover by about 1.5 percentage points (statistically significant at 10%).
- This is large relative to the unconditional forced turnover rate (the paper states it more than doubles it).
- This result supports the "forced turnover effect" as a reason to aim to meet/beat the target even if the pay schedule does not jump sharply at the target.

Table 3. Evidence for target ratcheting effect

Panel A: Previous performance and current goal levels

Goal	All (1)	EPS (2)	EPS (3)	Sales (4)	Sales (5)	Profit (6)	Profit (7)
Lag Performance	0.483***						
	(0.17)						
Lag EPS		1.08**	0.728*				
		(0.05)	(0.38)				
Lag Sale				0.761***	0.838***		
				(0.17)	(0.19)		
Lag Profits						0.440***	0.239*
						(0.13)	(0.12)
Size			0.242**		−1.576*		4.202***
			(0.97)		(0.92)		(0.79)
Constant	3.601	−0.616	−2.16**	−0.187	0.111	0.514***	−0.304***
	(2.98)	(1.13)	(0.92)	(1.43)	(0.69)	(0.07)	(0.07)
Observations	2,852	1,289	1,289	505	505	1,058	1,058
R^2	0.681	0.350	0.384	0.746	0.750	0.435	0.491

Panel B: Performance goals and actual performance

	Meet Target (1)	Meet Target (2)	Meet Target (3)	Meet Target (4)
Exceed target	0.095**	0.088**	0.108**	0.105**
	(0.045)	(0.044)	(0.053)	(0.052)
Size		−0.002		−0.01
		(0.013)		(0.014)
ROA		0.650***		0.378
		(0.237)		(0.277)
Sales growth		0.443***		0.310**
		(0.121)		(0.137)
Const.	0.543***	0.460***	0.558***	0.585***
	(0.013)	(0.120)	(0.015)	(0.132)
Obs.	1,954	1,954	1,395	1,395
R^2	0.210	0.227	0.211	0.219

Table 4. Performance goals and CEO turnover

	Forced (1)	Forced (2)
Miss target	0.015*	0.015*
	(0.008)	(0.008)
Target	0.009	0.011
	(0.007)	(0.007)
Target × Target	0.002	0.002
	(0.004)	(0.004)
Industry return	−0.015**	−0.028**
	(0.007)	(0.013)
Return		0.012
		(0.014)
Size		0.009**
		(0.004)
Volatility		0.205**
		(0.080)
Const.	−0.0005	−0.101**
	(0.007)	(0.046)
Obs.	1,874	1,874
R^2	0.354	0.360

Table 5. Univariate comparison of firms that exceed and miss performance goals

Panel A: EPS goals

	Exceed EPS goal		Miss EPS goal		Difference
	N	Mean	N	Mean	Exceed − Miss
Size	239	8.747	265	8.799	−0.052
ROA	239	0.110	265	0.111	−0.001
Market to book	239	1.811	265	1.809	0.002
Leverage	238	0.253	265	0.250	0.003
Sales growth	237	0.040	265	0.054	−0.014
Accruals	201	0.012	225	0.009	0.003
Repurchase	239	13.703	265	18.709	−5.006*
$\Delta R\&D$	239	0.743	265	1.839	−1.096**
$\Delta SG\&A$	239	9.523	265	9.249	0.274

Panel B: Sales goals

	Exceed Sales goal		Miss Sales goal		Difference
	N	Mean	N	Mean	Exceed − Miss
Size	171	8.996	231	9.092	−0.096
ROA	171	0.095	231	0.103	−0.008
Market to book	166	1.800	225	1.847	−0.047
Leverage	168	0.251	229	0.222	0.029
Sales growth	171	0.082	231	0.050	0.032**
Accruals	148	0.008	192	0.009	−0.001
Repurchase	171	16.237	231	16.613	−0.376
$\Delta R\&D$	171	2.906	231	1.766	1.140
$\Delta SG\&A$	171	11.880	231	11.026	0.854

Panel C: Profit goals

	Exceed Profit goal		Miss Profit goal		Difference
	N	Mean	N	Mean	Exceed − Miss
Size	129	9.004	228	8.728	0.276*
ROA	129	0.081	228	0.084	−0.003
Market to book	126	1.617	222	1.636	−0.019*
Leverage	129	0.287	228	0.273	0.014
Sales growth	128	0.019	228	0.052	−0.033**
Accruals	104	0.007	186	0.002	0.005
Repurchase	129	12.632	228	12.772	−0.140
$\Delta R\&D$	129	0.076	228	0.674	−0.598
$\Delta SG\&A$	129	−0.129	228	5.889	−6.018**

Interpretation of Table 5.

- EPS goals (Panel A): firms that barely exceed EPS goals look similar to firms that barely miss, *except* they (i) repurchase fewer shares and (ii) have a smaller change in R&D (more negative/less positive), consistent with cutting R&D to hit EPS.
- Sales goals (Panel B): the main difference is higher sales growth among firms that barely beat sales goals (mechanically consistent).
- Profit goals (Panel C): firms that barely exceed profit goals have notably lower changes in SG&A than firms that miss. This matches "real activities" manipulation via discretionary expense cuts.

Table 6. Multivariate difference between firms that exceed and miss performance goals

	Accruals (1)	$\Delta R\&D$ (2)	$\Delta SG\&A$ (3)
Exceed EPS	0.009**	−1.009*	
	(0.004)	(0.590)	
Exceed profit			−6.22**
			(3.016)
Size	0.002	−0.046	−0.549
	(0.002)	(0.233)	(0.879)
Market to book	0.0009	2.637**	11.762***
	(0.002)	(1.090)	(2.177)
Std. dev. cash flow	−0.018		
	(0.044)		
Std. dev. sales growth	-8.44×10^{-7}		
	(9.97×10^{-7})		
Const.	−0.016	−2.81	2.606
	(0.021)	(3.380)	(9.782)
Obs.	904	1,111	1,111
R^2	0.139	0.068	0.231

Interpretation of Table 6.

- Firms that barely exceed EPS goals have significantly higher abnormal accruals (0.009**) than firms that barely miss.
- Firms that barely exceed EPS goals also show lower changes in R&D (coefficient −1.009*), consistent with cutting R&D to hit EPS.
- Firms that barely exceed profit goals reduce SG&A more (coefficient −6.22**), consistent with discretionary expense cuts to boost profits.
- Overall: evidence points to both accrual-based and real-activity channels for "just beating" compensation targets.

7 Appendix B tables (examples of performance-linked grants) + interpretation

These tables are not central to the econometrics, but they help me visualize the shape of payout schedules and where kinks/steps can come from.

Table A.1. Barnes & Noble payout level (Example 1)

Level of Achievement of Consolidated Adjusted EBITDA Target	% of Target Payout
0% – less than 50%	0
50% – less than 75%	0.25
75% – less than 100%	0.625
100% – less than 112.5%	1
112.5% – less than 125%	1.085
125% or more	1.17

Interpretation of Table A.1. This is a clean, piecewise payout schedule with discrete slope changes (kinks) and a zero-payout region below threshold. It illustrates why I should expect strong incentives to avoid falling just short of threshold.

Table A.2. Quanta payout scale (Example 3)

Percentage of Operating Income Goal Attained	Payout as a Percentage of AIP Target Incentive
Less than 75%	0
0.75	0.25
0.80	0.40
0.85	0.55
0.90	0.70
0.95	0.85
1.00	1.00
1.10	1.30
1.20	1.75
1.30	1.85
1.40	1.95
150% or greater	2.00

Interpretation of Table A.2. This is a mostly smooth/interpolated schedule (dense grid) that can still embed convexities/concavities depending on how the slope changes across ranges.

Table A.3. Sunoco performance elements and weights (Example 4)

Incentive Plan Elements	Weight
Base Earnings per share	0.6
Revenue growth	0.2
Working capital improvement	0.2

Interpretation of Table A.3. This shows a classic multi-metric plan with weights. It helps explain why the paper predicts (and finds) less "just-barely beat" behavior when multiple metrics matter: it is harder to fine-tune all at once.

Interpretation of Table A.4. The thresholds/targets/maxima are explicit and the realized performance can land near these cutoffs. This makes it intuitive why density bunching/discontinuities around goal points are informative about incentives and possible gaming.

Table A.4. Sunoco payout levels (Example 4)

	Threshold	Target	Maximum	Actual 2006 Performance
Base Earnings per Share				
Amount	1.89	1.98	2.12	2.13
Percent of Prior Year	1.000	1.048	1.122	1.127
Revenue (Excl. Acquisitions)				
Amount (millions)	3528.6	3652.1	3705.3	3648.4
Percent of Prior Year	1.000	1.035	1.050	1.034
Working capital – cash gap days				
Reduction from Prior Year	0	3.25 days	6.5 days	7.2 days

8 Short summary

- When compensation is explicitly tied to specific accounting targets, firms' *reported* performance is disproportionately likely to end up just above those targets.
- The bunching is strongest in settings where I would most expect goal-gaming: EPS metrics, single-metric plans, concave schedules, non-equity payouts.
- Two reinforcing forces: targets appear to ratchet with past performance, and missing targets raises forced turnover risk.
- The "how" evidence is consistent with both accrual manipulation (abnormal accruals) and real actions (cuts to R&D or SG&A) to clear the target.

9 Conclusion

9.1 Big picture intuition

The core idea is simple: once a compensation contract embeds *explicit* performance goals (threshold/target/maximum), the pay–performance relationship (PPR) typically has *kinks* (slope changes) and sometimes *jumps*. When realized (or reported) performance is near one of these goal points, executives have strong incentives to avoid landing just below it. If managers can influence reported accounting outcomes (via accruals or real actions), then the distribution of Actual − Goal should show *bunching* just above zero: many firms barely beat the goal, and comparatively fewer barely miss it.

This "just-barely beat" pattern is a signature of goal-induced distortions in reporting/real decisions (though pure effort could also contribute in principle). The paper argues that explicit goals can therefore carry a *dark side*: they may improve accountability and transparency, but they also create focal points that encourage short-term actions to clear the goal.

9.2 What the paper concludes (main empirical conclusions)

Using a large sample of disclosed performance goals in CEO incentive grants (Incentive Lab, matched to Compustat/CRSP), the paper concludes:

1. **Performance clusters just above compensation goals.** Across accounting-based contracts, there is a statistically detectable discontinuity in the density of Actual − Target at zero: there are disproportionately many firm-years that exceed the target by a small margin relative to those that fall short by a similar margin.

2. **The effect is strongest where theory predicts gaming should be easiest/most attractive.** The "bunching" is particularly acute:
   - for **earnings/EPS goals** (strongest in EPS),
   - when compensation depends on a **single metric** (harder to barely beat multiple positively correlated metrics simultaneously),
   - when the PPR around the goal is **concave-shaped** (little marginal reward to outperforming far beyond the goal, so aiming just above the goal is rational),
   - and for **non-equity (cash) payout** grants (consistent with less convexity and stronger goal-focal incentives).

3. **Relative-performance goals do not show the same "just beat" pattern.** For goals stated in terms of *relative ranks/percentiles* versus peers, the paper does not find the same discontinuity at zero. Intuition: it is much harder to fine-tune relative performance when peer outcomes are unknown until after the period ends, so accounting/real manipulation is less effective for barely clearing a relative target.

4. **Two contract-dynamic / governance forces help explain why managers care about hitting targets even absent a large pay jump at the target.**
   - **Target ratcheting:** targets are positively related to past performance, and firms that just beat targets are more likely to meet targets next period (consistent with targets being set based on prior outcomes and/or continued "goal management").
   - **Forced turnover:** CEOs who miss targets are more likely to experience forced turnover, even after controlling for performance flexibly. This creates strong incentives to avoid falling short of the target.

   (The paper also notes reference-dependent preferences as a possible channel, but does not test it directly.)

5. **How do firms "just beat"? Evidence is consistent with both accrual and real actions.** Firms that just exceed EPS goals have **higher abnormal accruals** and **lower R&D spending** than firms that just miss EPS goals; firms that just exceed profit goals have **lower SG&A spending** than firms that just miss profit goals. This aligns with classic earnings-management channels: accrual manipulation and cuts to discretionary expenditures to push accounting metrics over the goal line.

9.3 What this means (the paper's practical takeaway)

The paper's overarching conclusion is that **linking managerial compensation to explicit accounting targets can create measurable distortions in reported performance and potentially in real corporate decisions**. These distortions are an important *cost* of goal-based pay design.

In terms of contract design implications:

- Distortions can be reduced if firms **avoid sharp focal point targets** (e.g., using smoother pay schedules without discrete goal points).

- If targets must be specified (e.g., for valuation/expensing/disclosure reasons), the evidence suggests **relative performance goals** are less prone to "just beat" behavior than absolute accounting goals.

- Among absolute accounting goals, the evidence suggests **EPS targets** are especially prone to goal-induced manipulation, while sales/profit targets show weaker patterns.
- Plans with **concave kinks** and **cash payouts** appear more associated with "just-beat" clustering, so committees should be cautious about these features when using explicit targets.

9.4 Takeaway

Overall, the paper documents a robust empirical regularity: when executive pay contracts specify explicit accounting targets, realized reported performance is unusually likely to land *just above* those targets. The pattern is strongest in settings where barely beating the target is most feasible and most valuable (EPS, single-metric plans, concave target regions, and non-equity payouts), and it vanishes for relative-performance targets where manipulation is harder. Additional evidence suggests managers may care about targets not only because of direct payout rules, but also because targets ratchet upward with past performance and missing them increases forced-turnover risk. Finally, firms that barely beat EPS/profit targets exhibit behaviors consistent with earnings management (abnormal accruals) and real activities manipulation (cuts to R&D or SG&A), highlighting the potential myopia costs of target-based compensation.